

Double Integrals

ANSWER KEY



Iterated integrals over rectangles

- 1 Evaluate $\int_0^2 \int_0^3 (x^2 + y) dy dx$.
Inner: $\int_0^3 (x^2 + y) dy = 3x^2 + \frac{9}{2}$. Outer: $\int_0^2 (3x^2 + \frac{9}{2}) dx = [x^3 + \frac{9}{2}x]_0^2 = 8 + 9 = 17$.
- 2 Evaluate $\int_1^2 \int_0^1 6xy^2 dx dy$.
Inner: $\int_0^1 6xy^2 dx = 3y^2$. Outer: $\int_1^2 3y^2 dy = [y^3]_1^2 = 8 - 1 = 7$.

Double integrals over general regions

- 3 Evaluate $\int_0^1 \int_0^x (x + y) dy dx$, integrating over the triangular region $0 \leq y \leq x \leq 1$.
Inner: $\int_0^x (x + y) dy = x^2 + \frac{x^2}{2} = \frac{3x^2}{2}$. Outer: $\int_0^1 \frac{3x^2}{2} dx = [\frac{x^3}{2}]_0^1 = \frac{1}{2}$.
- 4 Use a double integral to find the volume under $f(x, y) = 4 - x - y$ over the rectangle $0 \leq x \leq 1, 0 \leq y \leq 1$.
 $\int_0^1 \int_0^1 (4 - x - y) dy dx$: inner $= 4 - x - \frac{1}{2}$. Outer: $\int_0^1 (\frac{7}{2} - x) dx = \frac{7}{2} - \frac{1}{2} = 3$.

Reversing the order of integration

- 5 Evaluate $\int_0^1 \int_0^1 e^{x^2} dx dy$ by first reversing the order of integration (the region is $0 \leq y \leq x \leq 1$).
Reversing: $\int_0^1 \int_0^x e^{x^2} dy dx = \int_0^1 x e^{x^2} dx = [\frac{1}{2} e^{x^2}]_0^1 = \frac{e-1}{2}$.

Double integrals in polar coordinates

- 6 Evaluate $\iint_D (x^2 + y^2) dA$ where D is the disk $x^2 + y^2 \leq 4$, by converting to polar coordinates ($dA = r dr d\theta$).
 $\int_0^{2\pi} \int_0^2 r^2 \cdot r dr d\theta = \int_0^{2\pi} [\frac{r^4}{4}]_0^2 d\theta = \int_0^{2\pi} 4 d\theta = 8\pi$.
- 7 Find the area of the region enclosed by $r = 2\cos\theta$ using a double integral in polar coordinates.
 $A = \int_{-\pi/2}^{\pi/2} \int_0^{2\cos\theta} r dr d\theta = \int_{-\pi/2}^{\pi/2} 2\cos^2\theta d\theta = \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta) d\theta = \pi$.

Average value over a region

- 8 Find the average value of $f(x, y) = xy$ over the rectangle $0 \leq x \leq 2, 0 \leq y \leq 3$ (average value $= \frac{1}{A} \iint_A f dA$, where A is the region's area).
 $\iint_D xy dA = \int_0^2 \int_0^3 xy dy dx = \int_0^2 x \cdot \frac{9}{2} dx = \frac{9}{2} \cdot 2 = 9$. Area $= 6$. Average $= \frac{9}{6} = \frac{3}{2}$.

Mass and center of a region

- 9 A lamina occupies the rectangle $0 \leq x \leq 2$, $0 \leq y \leq 1$ with density $\rho(x, y) = x + y$. Find its total mass,

$$M = \iint_R \rho \, dA.$$

$$M = \int_0^2 \int_0^1 (x + y) \, dy \, dx = \int_0^2 \left(x + \frac{1}{2}\right) dx = \left[\frac{x^2}{2} + \frac{x}{2}\right]_0^2 = 2 + 1 = 3.$$

- 10 For a lamina occupying $0 \leq x \leq 1$, $0 \leq y \leq 1$ with constant density $\rho = 1$, find the x -coordinate of the center of mass, $\bar{x} = \frac{1}{M} \iint_R x \, dA$ (where M is the total mass).

$$M = 1 \cdot 1 = 1. \quad \iint_D x \, dA = \int_0^1 \int_0^1 x \, dy \, dx = \int_0^1 x \, dx = \frac{1}{2}. \quad \bar{x} = \frac{1/2}{1} = \frac{1}{2} \text{ (as expected, the center of a uniform square).}$$

Triple integral setup (a first look)

- 11 Set up (do not evaluate) the triple integral for the volume of the solid bounded by $0 \leq x \leq 1$, $0 \leq y \leq 2$, $0 \leq z \leq 3$.

$$V = \int_0^1 \int_0^2 \int_0^3 dz \, dy \, dx \text{ (which evaluates to } 1 \times 2 \times 3 = 6, \text{ the box's volume).}$$